MIL-N-3667B
24 April 1984
SUPERSEDING
MIL-N-3667A
19 November 1968

MILITARY SPECIFICATION

NOZZLE, METEOROLOGICAL, BALLOON INFLATION ML-373/GM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of meteorological balloon inflation device designated Nozzle, Meteorological, Balloon Inflation ML-373/GM.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods Of.
MIL-M-13231	- Marking of Electronic Items.
MIL-F-14072	- Finishes For Ground Electronic Equipment.
MIL-P-19834	- Plate, Identification, Metal Foil, Adhesive Backed.
MIL-E-55585	- Electronic Equipment And Parts, Packaging Of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6660

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-169	- Extreme-Temperature Cycle.
MIL-STD-454	- Standard General Requirements For Electronic Equipment.
MIL-STD-726	- Packaging Requirements Code.
MIL-STD-810	- Environmental Test Methods.

2.1.2 Other Government documents. The following Government drawings form a part of this specification to the extent specified herein:

DRAWINGS

ELECTRONICS COMMAND

SC-D-15182	- Balloon Nozzle, ML-373/GM, Assembly And Details.
SC-D-16286	- Standard Cycle For Moisture Resistance Tests Of Component Parts.

(Copies of specifications, standards and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 Description. The nozzle consists of a two-way plug valve with one inlet for attachment of a .50 inch (12.7mm) inside diameter rubber hose and two different sized outlets for attachment of a balloon during inflation. The valve can be adjusted to connect the inlet to either outlet, or to shut off both outlets.

3.2 First article. When specified (see 6.2), a sample shall be subject to first article inspection (see 4.3 and 6.3).

3.3 Material. The material shall be as shown on the applicable drawings and as specified herein. However, when a definite material is not specified, a material shall be used which will enable the nozzles to meet the requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Cleaning. All cleaning of material, parts and equipment shall be in accordance with MIL-STD-454, requirement 9.

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3.3.2 Recycled, virgin, and reclaimed material. It is encouraged that recycled or reclaimed material, in lieu of virgin material be used when practical provided it meets the requirements of this specification.

3.4 Construction. The nozzle shall be constructed in accordance with Drawing SC-D-15182, and as specified herein.

3.5 Performance.

3.5.1 Nozzle. The nozzle shall withstand a pressure of 8 inches of mercury (in.Hg) (27kPa). The loss of pressure due to leakage shall be not greater than .50 in.Hg (1.7kPa) in 15 seconds for either closed position (see 4.6.1).

3.5.2 Pointer. The pointer of the handle shall be positioned on the valve so that the pointer indicates the injector being fed (see 4.6.1).

3.6 Environmental conditions. The nozzle, and assemblies shall withstand individual environmental conditions specified herein, or combinations thereof, without damage, performance degradation, or malfunction when tested in accordance with 4.6.2.

3.6.1 Operation.

a. Temperature. Any ambient temperature in the range of -60° to 120°F (-51.1°C to 48.9°C).

b. Relative humidity. Any relative humidity up to 100 percent.

c. Salt fog. Exposure for not less than 48 hours.

3.7 Lubrication. The plug valve shall be lubricated with a light coat of special purpose lubricating oil, Cities Service Company, Tulsa, OK p/n 3908, or equal. Any excess lubricant shall be removed from the gas passage ways.

3.8 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the nozzle. Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, or characteristic.

3.9 Weight. The weight of each nozzle and nozzle weight shall be as shown on Drawing SC-D-15182.

3.10 Finish. The nozzle shall be finished in accordance with MIL-F-14072 and the equipment drawings.

3.11 Marking. The nameplate material shall be in accordance with MIL-P-19834. Marking format shall be in accordance with MIL-M-13231.

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3.12 Workmanship. The equipment shall be manufactured and assembled in accordance with MIL-STD-454, requirement 9 and the applicable portions of the following paragraphs herein:

3.3.1	Cleaning
3.4	Construction
3.7	Lubrication
3.10	Finish

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Inspection covered by subsidiary documents (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Packaging inspection (see 4.7).

4.3 First article.

4.3.1 First article units. The contractor shall furnish five first article units of the complete equipment.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4 and the inspections specified for group A and group C (see tables I and II). The inspection shall be performed in the following order: 4.4, group A for all units; and group C in the order shown in table II. After completion of group C tests, conforming units shall be reinspected and shall pass all group A inspections. Failure of any inspection shall be cause for rejection.

4.4 Inspection covered by subsidary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification.

Item	Where required
Finish	3.10
Marking	3.11

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4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4 and 4.5.1 through 4.5.3.3.

4.5.1 Group A inspection. Each unit shall be inspected for conformance to the inspections specified in table I. Discrete lots shall be formed from units that pass this inspection. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and acceptable quality levels (AQL) indicated in table I.

TABLE I. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical	3.12	4.6.5	1.0	4.0
Interchangeability	3.8	4.6.3	4.0	6.5
Weight	3.9	4.6.4	1.0	4.0

4.5.2 Group B inspection. Group B inspection not required.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A inspection. The inspection shall consist of the inspections specified in table II. Samples shall be selected in accordance with 4.5.3.1.

TABLE II. Group C inspection.

Inspection	Requirement paragraph	Inspection paragraph
Temperature	3.6.1a	4.6.2.1
Humidity	3.6.1b	4.6.2.2
Fog	3.6.1c	4.6.2.3 1/
Leak	3.5.1	4.6.1

1/ This test shall be performed on a separate sample.

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4.5.3.1 Group C sampling plan. Five sample units shall be selected at rando each group C inspection in table II from the first 50 units produced, and five selected at random from each 100 or fraction thereof produced. When first a samples are required by the contract or purchase order, the testing of five units the first 100 production units shall be omitted.

4.5.3.2 Group C failures. Actions relative to group C failures shall be as speci the contract (see 6.2).

4.6 Methods of inspection.

4.6.1 Leak test. Each nozzle shall be subjected to a pressure of 8 in. Hg (using a manometer or similar equipment. Connection between manometer and shall be as short as possible. Test shall be made with valve in closed position ar rotated 180° to opposite closed position. Nonconformance to 3.5.1 and 3.5. constitute failure of this test.

4.6.2 Environmental test.

4.6.2.1 Temperature. The equipment shall be tested in accordance with MI 169, steps 1, 4, 5, 7, 8, 9 and 10. Steps 2, 3 and 6 shall be omitted. The equipme then be resubjected to 4.6.1 and nonconformance to 3.5.1 and 3.5.2 shall co failure of this test.

4.6.2.2 Moisture-resistance. The nozzle shall be subjected to 10 continuous c the moisture-resistance test shown on Drawing SC-D-16286. The leak tes 4.6 be performed prior to, after 5 cycles of, and at the end of this humidity tes vibration test specified on Drawing SC-D-16286 shall be omitted. Nonconform 4.6.1 shall constitute failure of this test.

4.6.2.3 Salt-fog. The equipment and assemblies shall be exposed in their o configuration to salt fog in accordance with MIL-STD-810, method 509.1 for 4 Equipment and assemblies shall not be opened and shall not be required to be o The equipment shall then be resubjected to 4.6.1 and nonconformance to 3.5.1 a shall constitute failure of this test.

4.6.3 Inspection for interchangeability. The dimensions shall be gaged or n to determine compliance with the physical interchangeability requirements When a dimension is not within specified or design limits, it shall be considered defect.

4.6.4 Weight test. Each nozzle and nozzle weight shall be Nonconformance to Drawing SC-D-15182 shall constitute failure of this test.

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4.6.5 Visual and mechanical. The nozzle shall be inspected for conformance to 3.12 and 4.4. Construction and physical dimensions shall be inspected for conformance to the applicable drawings and as specified herein.

4.7 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

4.7.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.7.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.7.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table III, "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE III. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied, or omitted.	Visual
109	Marking omitted, incorrect, or illegible.	Visual

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5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 2M-1-1-09-GH-NU-X-XX-A-ED-A.

5.1.2 Level B. 2M-1-1-09-GH-NU-X-XX-A-ED-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The Nozzle, Meteorological, Balloon Inflation ML-373/GM is used in the inflation and "weighing-off" of 30 and 100-gram meteorological balloons.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval (see 3.2).
c. Action required relative to group C failures (see 4.5.3.2).
d. Level A or B preservation and packing (see section 5).
e. When first article inspection rough handling tests are not required.
f. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction sample. The first article should consist of five units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.5 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ER
Air Force - 99

Review activity:
Air Force - 80
DLA - GS

Preparing activity:
Army - ER

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-N-3667B	NOZZLE, METEOROLOGICAL, BALLOON INFLATION ML-373/GM

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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